

Rice

Oryza sativa — Pakistan's Premier Export Cereal

A Complete Cultivation Guide for

Punjab · Sindh · Khyber Pakhtunkhwa · Balochistan

Compiled for Farmer Guidance & AI Advisory Use

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Table of Contents

1. Introduction
2. Scientific Name
3. Importance in Pakistan
4. Major Producing Provinces and Districts
5. Climate Requirements
6. Soil Requirements
7. Land Preparation
8. Recommended Varieties (Pakistan)
9. Seed Rate
10. Seed Treatment
11. Sowing Time (Province-wise)
12. Sowing Methods
13. Fertilizer Recommendations (NPK)
14. Irrigation Schedule
15. Weed Management
16. Insect Pests
17. Diseases
18. Deficiency Symptoms
19. Harvesting
20. Storage
21. Expected Yield in Pakistan
22. Government Recommendations
23. Modern Farming Practices
24. Precision Agriculture
25. Sustainable Farming
26. Regional Notes

1. Introduction

Rice is one of Pakistan's most important staple and export crops, grown mainly in the irrigated plains of Punjab and Sindh during the summer (kharif) season. Pakistan is globally known for its fine, aromatic Basmati rice, which commands premium prices in export markets, alongside coarser IRRI-type varieties grown for domestic consumption and bulk export. Rice cultivation is water-intensive and traditionally grown under standing water (puddled transplanted culture), though water scarcity is pushing farmers and researchers toward more efficient methods. This guide explains how rice is grown across Pakistan's rice-growing provinces and how farmers can improve yield, quality, and water-use efficiency.

2. Scientific Name

Rice's scientific name is *Oryza sativa*, belonging to the Poaceae (grass) family. Pakistan grows two broad types: Basmati rice (long-grain, aromatic, mainly for export) and coarse/IRRI-type rice (higher yielding, shorter grain, mainly for domestic use and bulk export).

3. Importance in Pakistan

Rice is Pakistan's second-largest export crop after cotton textiles in terms of agricultural export earnings, with Basmati rice in particular building the country's international reputation for aromatic long-grain rice. It is grown on roughly 2.5–3 million hectares and is a staple food for much of the population, especially in rice-growing regions of Punjab and Sindh where it is a dietary staple alongside wheat. Rice cultivation also supports millions of livelihoods through farming, milling, and export trade, making it a crop of major economic significance for both farm incomes and national foreign exchange earnings.

4. Major Producing Provinces and Districts

- Punjab: Sheikhpura, Gujranwala, Hafizabad, Sialkot, Narowal, and the Kalar tract — Pakistan's premier Basmati-growing belt known worldwide for quality aromatic rice.
- Sindh: Larkana, Shikarpur, Dadu, Thatta, and the lower Indus tract — major producer of coarse and IRRI-type rice, along with some Basmati.
- Khyber Pakhtunkhwa: Swat, Mardan, and Peshawar valley — smaller but locally important rice-growing areas.
- Balochistan: Limited cultivation in Jaffarabad and Nasirabad districts where irrigation water is available.

5. Climate Requirements

Rice is a warm-season crop requiring temperatures between 25°C and 35°C during the growing period, with high humidity and abundant water availability being essential, especially during the traditional puddled transplanted system. Basmati varieties in particular need cooler nights during grain-filling and ripening (typically in September–October in Punjab) to develop their characteristic aroma and grain quality, which is one reason Punjab's Kalar tract has historically produced superior Basmati compared to hotter regions.

6. Soil Requirements

Rice grows best in heavy clay or clay loam soils that can hold standing water well, since the traditional puddled transplanting system relies on a compacted, water-retentive soil layer to reduce percolation losses. A soil pH between 5.5 and 7.5 is generally suitable, and rice can tolerate somewhat saline and waterlogged conditions better than most other cereal crops, making it a useful crop for reclaiming certain waterlogged and salt-affected lands in Punjab and Sindh.

7. Land Preparation

Fields are first ploughed and then flooded and puddled (repeated ploughing and harrowing under standing water) to create a compact, impervious layer that reduces water seepage and suppresses weeds, a practice central to traditional rice cultivation in Pakistan. Basmati fields in particular are carefully leveled using laser land leveling where available, since even water distribution is critical for uniform crop stand and reduced water wastage.

8. Recommended Varieties (Pakistan)

- Basmati types: Super Basmati, Basmati-515, Basmati-2000, and Kissan Basmati are among the widely grown varieties recommended by the Rice Research Institute Kala Shah Kaku for Punjab's premium export market.
- Coarse/IRRI types: IRRI-6 and IRRI-9 are popular high-yielding varieties grown extensively in Sindh for both domestic consumption and bulk export.
- Hybrid rice varieties are increasingly promoted, particularly in Punjab and Sindh, to boost yield beyond what conventional varieties achieve, though Basmati remains dominant for the premium export segment due to its unique aroma and grain elongation on cooking.
- Farmers should confirm the latest variety recommendations each season from the Rice Research Institute and provincial seed corporations, since new higher-yielding and disease-resistant Basmati lines are periodically released.

9. Seed Rate

For transplanted rice, a nursery seed rate of about 8–10 kg per acre is recommended to raise enough seedlings for transplanting into one acre of main field. For direct-seeded rice (an increasingly promoted water-saving method), a higher seed rate of about 12–15 kg per acre is used since there is no nursery transplanting step to select the strongest seedlings.

10. Seed Treatment

Seed should be treated with a recommended fungicide such as carbendazim or thiram (2–3 g per kg seed) before sowing in the nursery to protect against seedling blight and other seed-borne diseases. Seed should also be soaked in water for 24 hours and then incubated for sprouting before sowing, a traditional practice that improves germination uniformity in the nursery bed.

11. Sowing Time (Province-wise)

- Punjab: Nursery sown late May to early June; transplanting done in June–July, timed so that Basmati varieties mature and ripen during the cooler September–October period for best grain quality and aroma.
- Sindh: Nursery sown May–June; transplanting in June–July, slightly earlier than Punjab due to the province's overall warmer climate.
- Khyber Pakhtunkhwa: Nursery sown late May; transplanting in June, adjusted to local valley temperatures, particularly in Swat.
- Balochistan: Sown June where irrigation water is available, though cultivation area is limited compared to Punjab and Sindh.

12. Sowing Methods

The traditional and still dominant method in Pakistan is transplanting 25–35 day old nursery seedlings into puddled, flooded fields at a spacing of about 20 x 15 cm, generally two to three seedlings per hill. Direct seeding of rice (dry or wet-seeded), which saves water and labor by skipping the nursery and transplanting steps, is being increasingly promoted, especially in parts of Punjab facing water and labor shortages, though it requires more careful weed management since standing water cannot be used as early to suppress weeds.

13. Fertilizer Recommendations (NPK)

A general recommendation for Basmati rice is 80–100 kg Nitrogen, 60 kg Phosphorus (P_2O_5), and 50–60 kg Potash (K_2O) per acre, while coarse/IRRI varieties can utilize somewhat higher nitrogen (up to 120–130 kg per acre) given their greater yield potential and lodging resistance. Phosphorus and potash and about one-third of nitrogen are applied at transplanting or before final puddling, with the remaining nitrogen split into two or three top-dressings at active tillering, panicle initiation, and heading stages.

14. Irrigation Schedule

Traditional rice cultivation in Pakistan maintains 5–7 cm of standing water in the field throughout most of the growing season, with water drained a couple of weeks before harvest to allow the field to firm up for easier harvesting. Because rice is Pakistan's most water-intensive crop, water-saving techniques such as alternate wetting and drying (allowing the field to dry briefly between irrigations) and direct seeding are being actively promoted by research institutes to reduce water use without significantly sacrificing yield.

15. Weed Management

Standing water in traditionally transplanted rice provides significant natural weed suppression, though grassy and sedge weeds can still emerge and compete for nutrients if water management is inconsistent. In direct-seeded rice, weed pressure is considerably higher since early flooding is not always possible, making pre-emergence and post-emergence herbicides, along with timely hand weeding, more critical for maintaining yield in this system.

16. Insect Pests

Major insect pests of rice in Pakistan include the rice stem borer (causing "dead heart" in young plants and "white ear" in mature ones), rice leaf folder, brown planthopper, and rice hispa. Stem borer and planthopper populations can build up quickly under continuous rice cultivation and excessive nitrogen use, so integrated pest management combining resistant varieties, balanced fertilization, natural predator conservation, and need-based insecticide application is recommended over routine calendar spraying.

17. Diseases

Rice blast (a fungal disease affecting leaves, nodes, and panicles, capable of causing severe yield loss under humid conditions) and bacterial leaf blight are the most economically significant diseases of rice in Pakistan. Sheath blight and brown spot can also occur, particularly under dense planting and high nitrogen conditions. Management includes resistant varieties, balanced (not excessive) nitrogen fertilization, proper field drainage, and fungicide application where blast or bacterial blight pressure is confirmed to be high.

18. Deficiency Symptoms

Nitrogen deficiency shows as overall pale yellow-green leaves and reduced tillering, directly impacting the number of productive panicles. Phosphorus deficiency causes stunted growth, delayed maturity, and darker green, sometimes purplish leaves. Potassium deficiency appears as yellowing and browning along leaf margins along with weak stems prone to lodging. Zinc deficiency is a well-known and widespread problem in Pakistan's rice soils, especially under continuous flooding, showing as stunted growth and brown blotching on leaves (khaira disease), and is typically corrected with zinc sulfate application.

19. Harvesting

Rice is ready for harvest when about 80–85% of the grains in the panicle turn golden yellow and the moisture content drops to a suitable level, typically 100–150 days after transplanting depending on variety, with Basmati varieties generally taking longer to mature than coarse IRRI types. Harvesting is done by hand with sickles in much of Punjab's traditional Basmati belt to preserve grain quality, though combine harvesters are increasingly used for coarse varieties in Sindh and parts of Punjab to reduce labor costs.

20. Storage

Harvested paddy (unmilled rice) must be dried to a safe moisture level (around 12–14%) before storage or milling to prevent mold growth and maintain milling quality, since high-moisture paddy is prone to yellowing and breakage during milling. Paddy is typically stored in jute bags or bulk storage facilities in cool, dry, well-ventilated conditions, with rice millers and exporters investing in proper drying and storage infrastructure to preserve the premium quality that Basmati rice commands in export markets.

21. Expected Yield in Pakistan

The national average paddy yield in Pakistan is around 2.5–3 tons per hectare for Basmati varieties and somewhat higher, around 3.5–4.5 tons per hectare, for coarse/IRRI varieties, reflecting the yield-quality tradeoff between premium aromatic rice and higher-yielding bulk varieties. Progressive farmers using hybrid or improved varieties, balanced fertilization, zinc correction, and better water management in Punjab and Sindh can achieve significantly higher yields than these national averages.

22. Government Recommendations

The Rice Research Institute at Kala Shah Kaku (Punjab) and the Rice Research Institute Dokri (Sindh) lead variety development and agronomic research for the crop, working alongside provincial extension departments to promote certified seed use, zinc application, and water-saving technologies. Government policy has historically supported Basmati export promotion given its importance to foreign exchange earnings, including quality standards and branding efforts to protect Pakistan's Basmati reputation in international markets.

23. Modern Farming Practices

Laser land leveling, direct seeding of rice to save water and labor, and mechanical transplanters are increasingly being adopted on larger and more progressive farms in Punjab and Sindh. Combine harvesting for coarse varieties and improved drying and storage facilities at the mill level are also modernizing the post-harvest side of the rice value chain, helping reduce losses and improve export quality consistency.

24. Precision Agriculture

Precision practices being introduced in rice include soil and water testing to guide zinc and other micronutrient applications, alternate wetting and drying irrigation scheduling guided by field water-level indicators to save water without yield loss, and satellite-based crop area and health monitoring used by some research institutes and large commercial operations to track crop progress across large rice-growing tracts.

25. Sustainable Farming

Because rice is Pakistan's most water-intensive major crop, sustainability efforts focus heavily on reducing water use through alternate wetting and drying, direct seeding, and laser leveling, all of which can substantially cut water requirements without proportional yield loss. Reducing stubble burning after harvest (a widespread but environmentally harmful practice that contributes to air pollution) in favor of proper residue management or incorporation is another major sustainability priority being promoted by government and research institutes in Punjab.

26. Regional Notes

Punjab's Basmati belt, centered on the Kalar tract, remains the heart of Pakistan's premium rice export industry, benefiting from cooler autumn nights that develop the crop's signature aroma, while Sindh's rice

areas focus more on higher-yielding coarse and IRRI varieties for bulk consumption and export. As water scarcity becomes an increasingly pressing issue across both provinces, the shift toward water-saving cultivation methods like direct seeding and alternate wetting and drying is likely to become progressively more important for the long-term sustainability of Pakistan's rice sector.